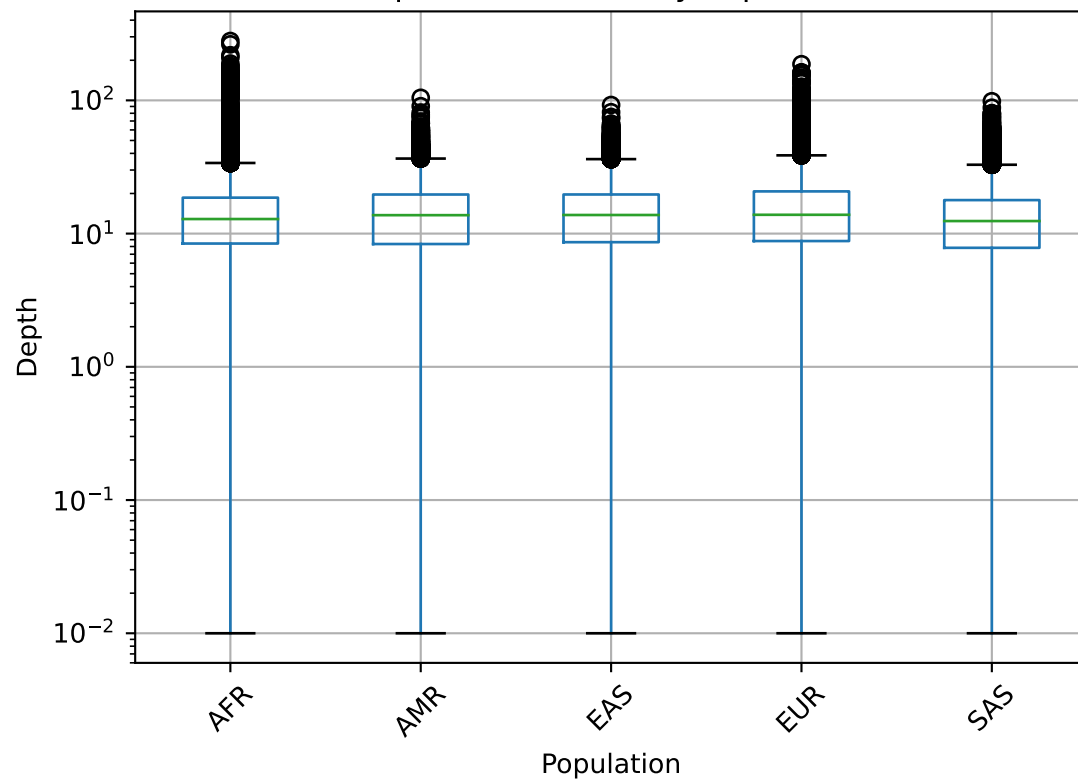
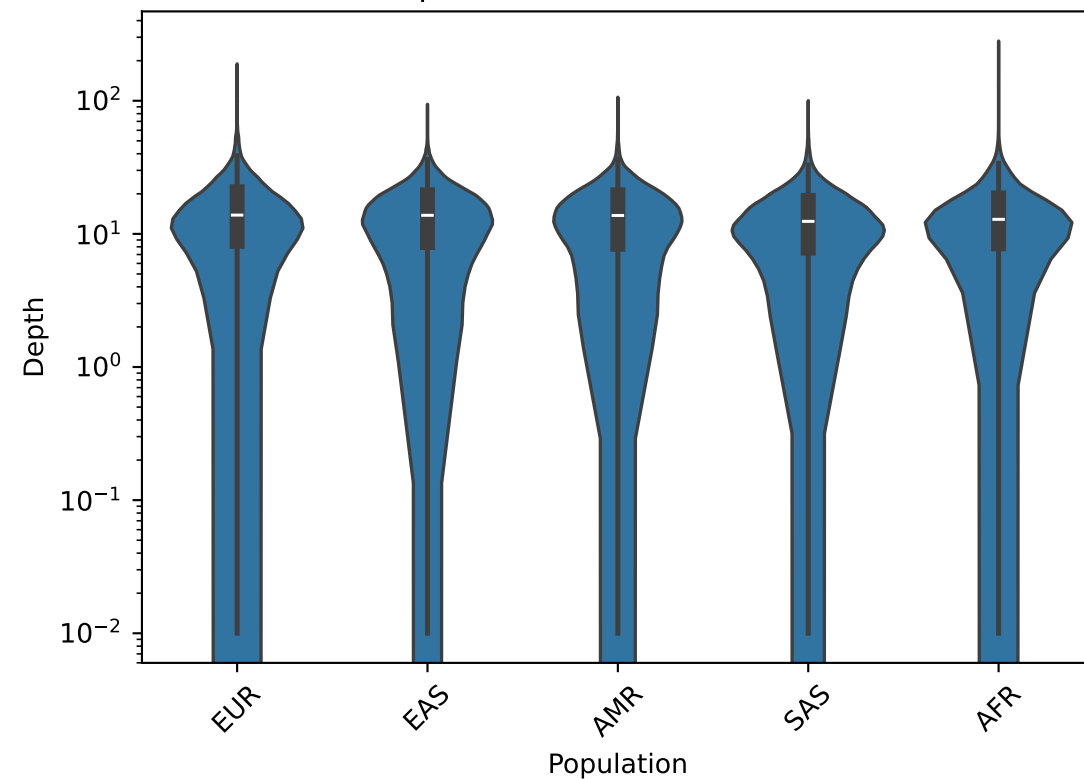


Boxplot groupedby super_pop

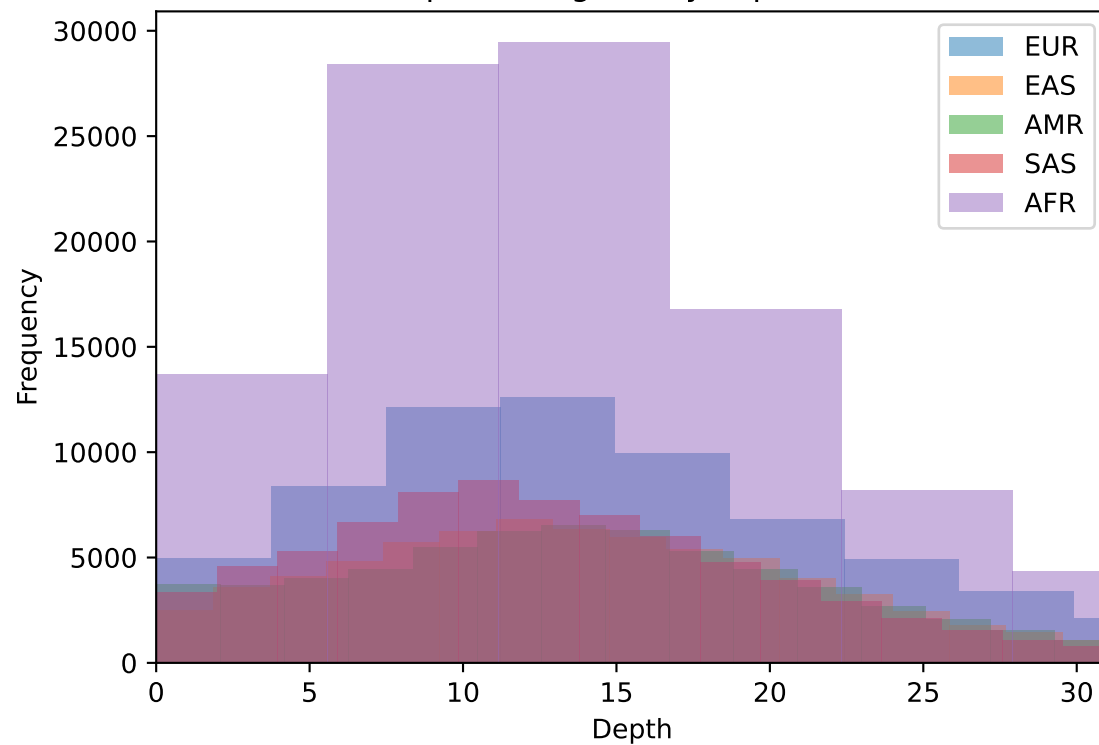
Depth Distribution by Population



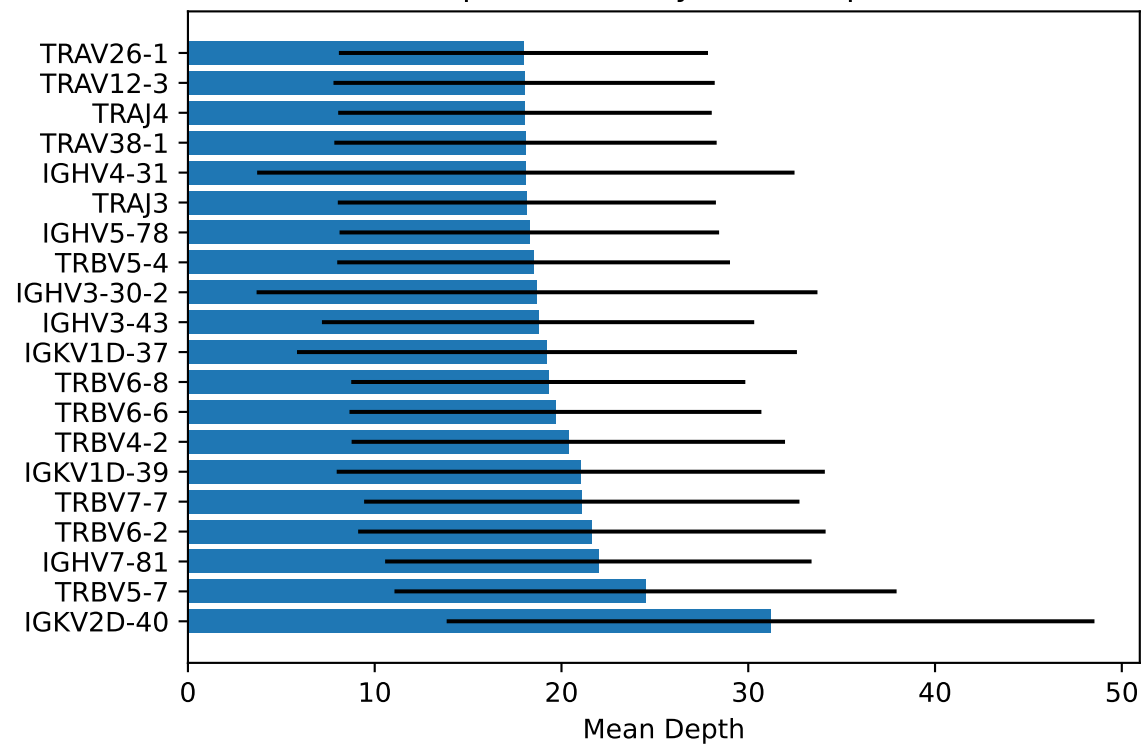
Depth Distribution (Violin Plot)



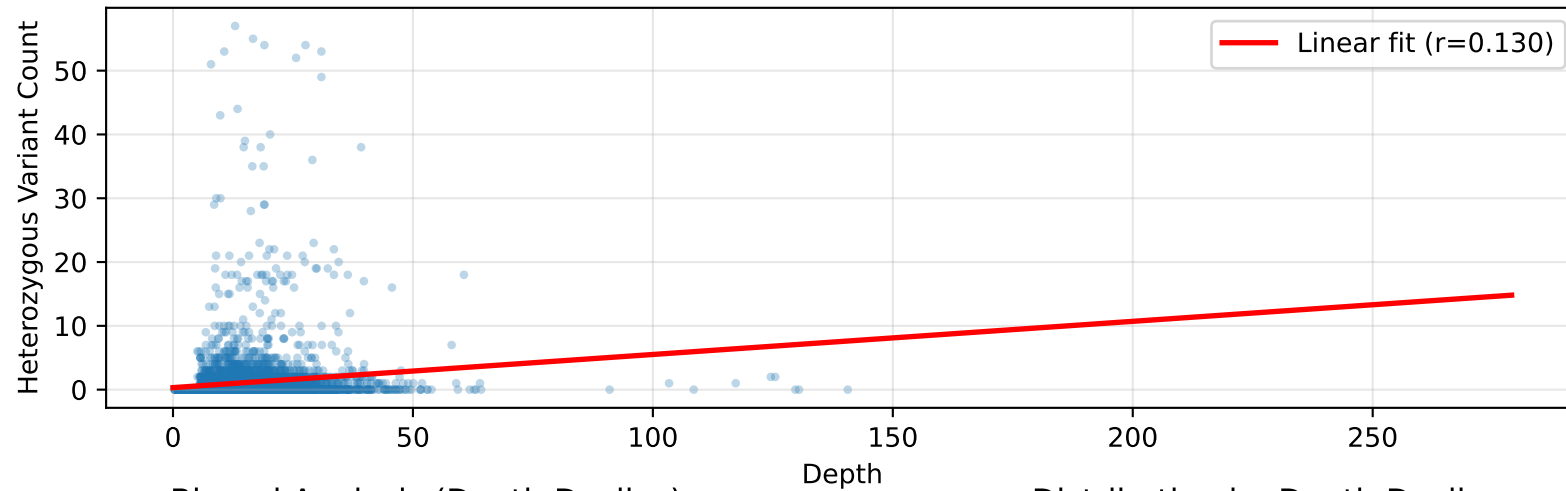
Depth Histogram by Population



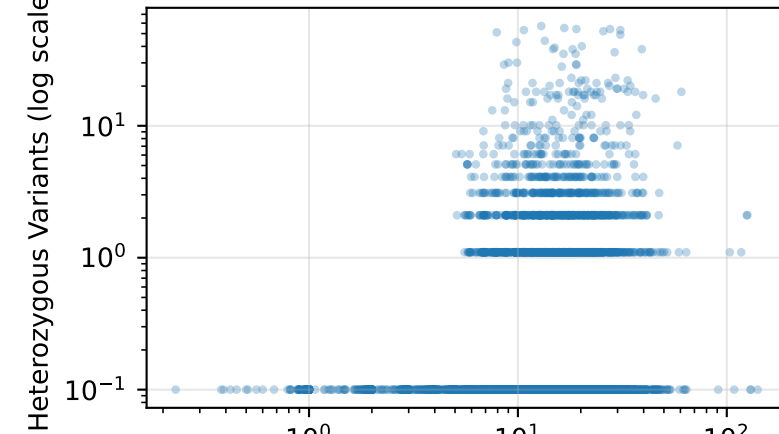
Top 20 Genes by Mean Depth



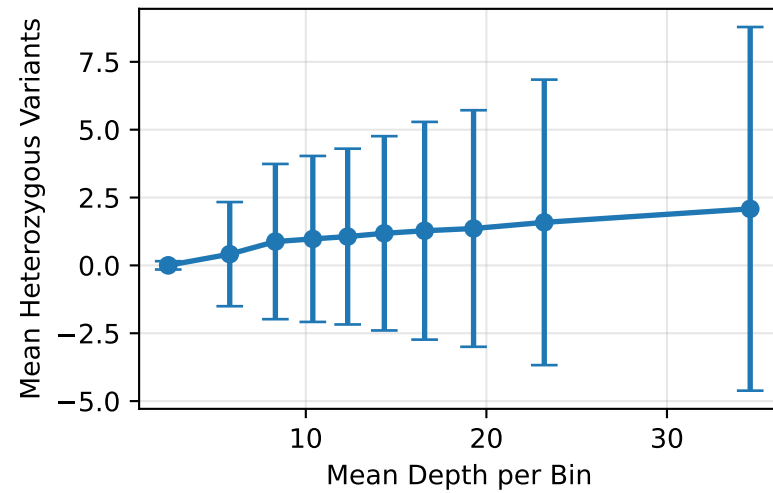
Boxplot grouped by depth_decile

Depth vs Heterozygosity
Pearson $r=0.1297$ ($p=0.00e+00$), Spearman $\rho=0.2227$ ($p=0.00e+00$)

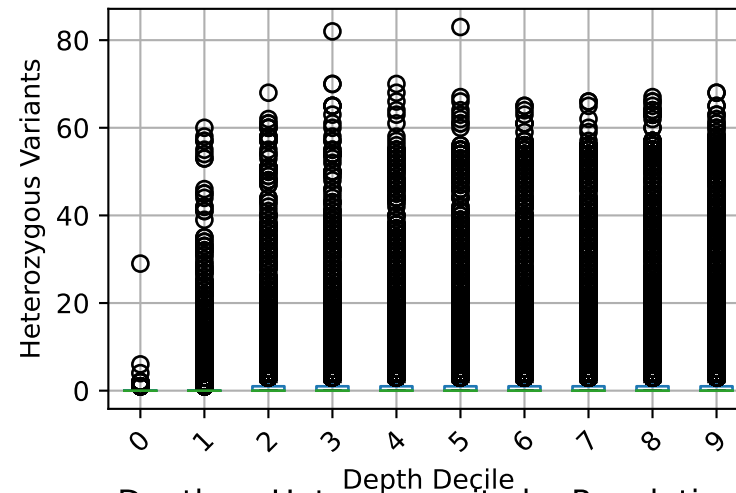
Log-Log Relationship



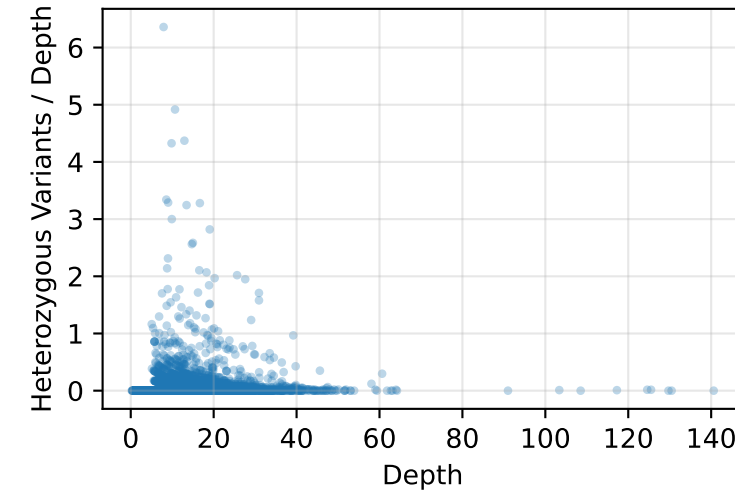
Binned Analysis (Depth Deciles)



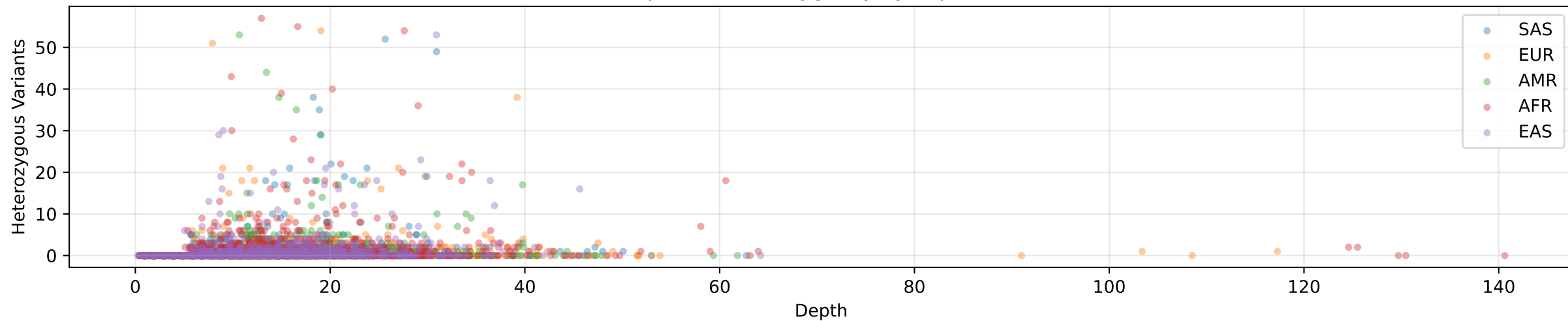
Distribution by Depth Decile



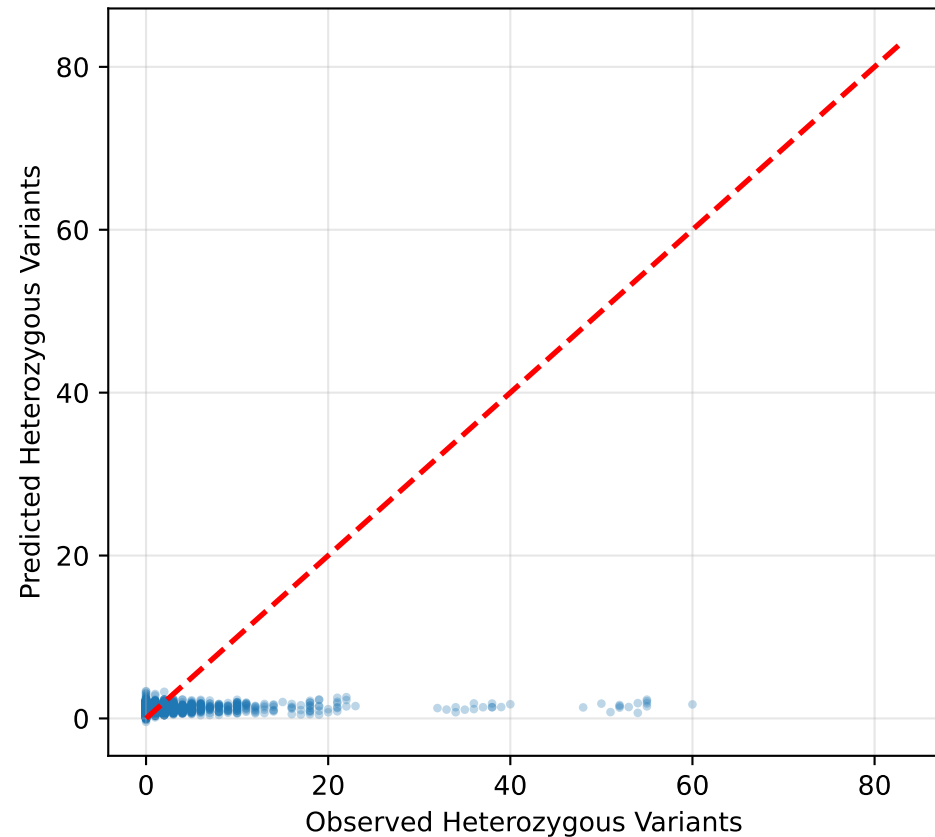
Normalized Heterozygosity Rate



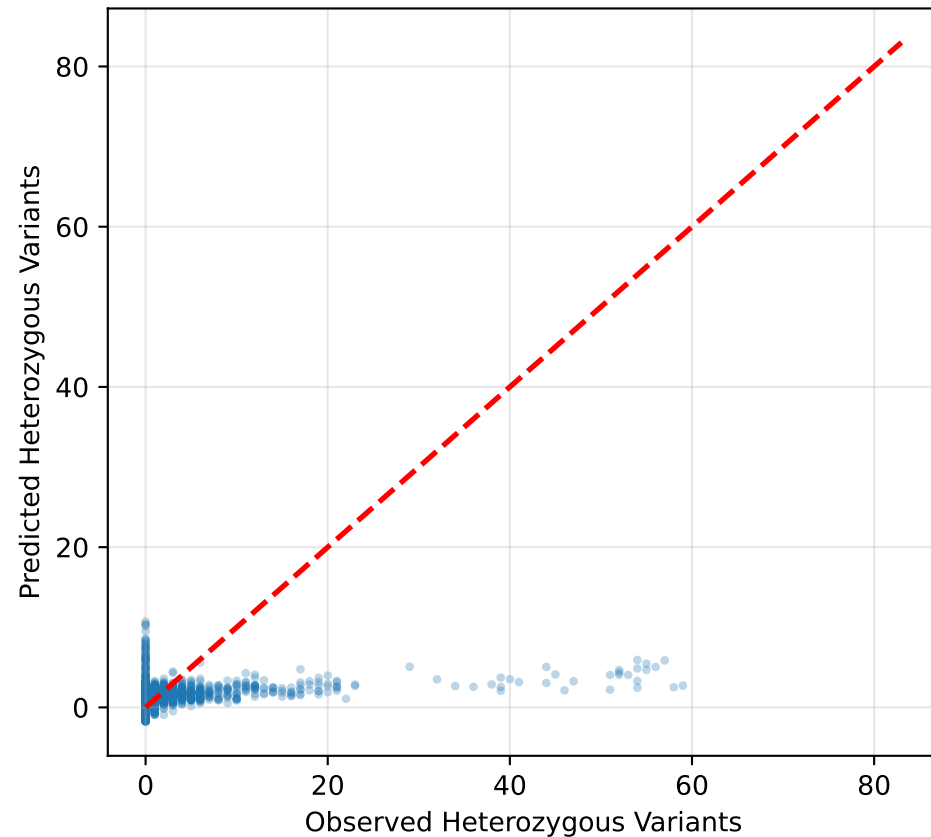
Depth vs Heterozygosity by Population



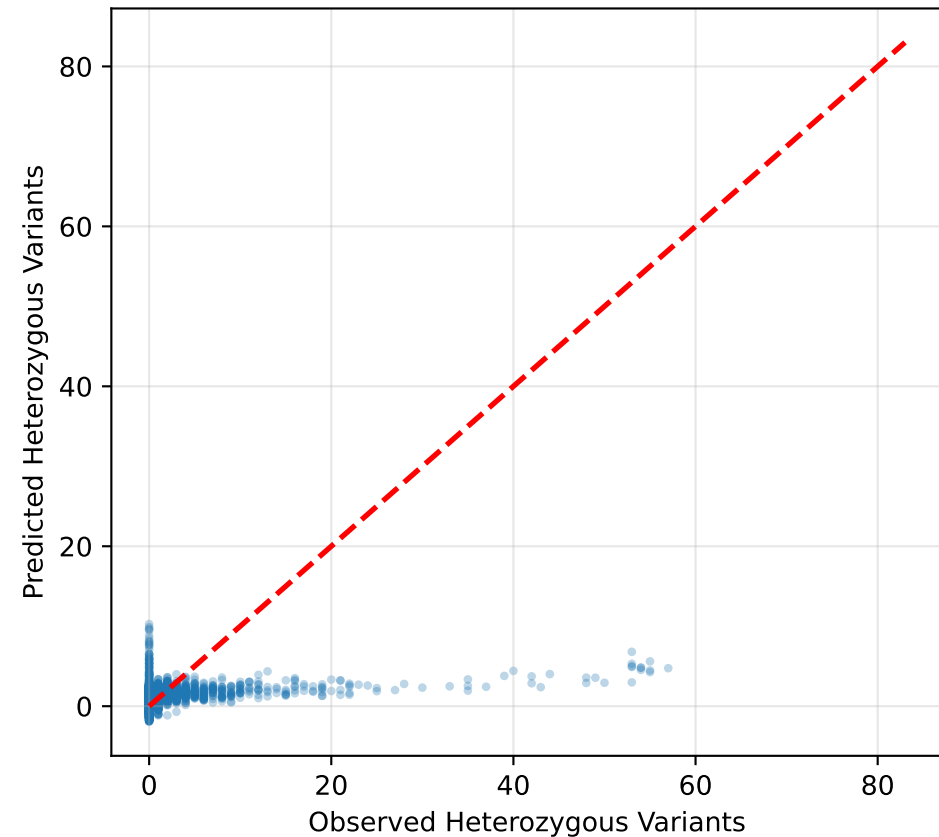
Model 1: Depth only
 $R^2 = 0.0141$



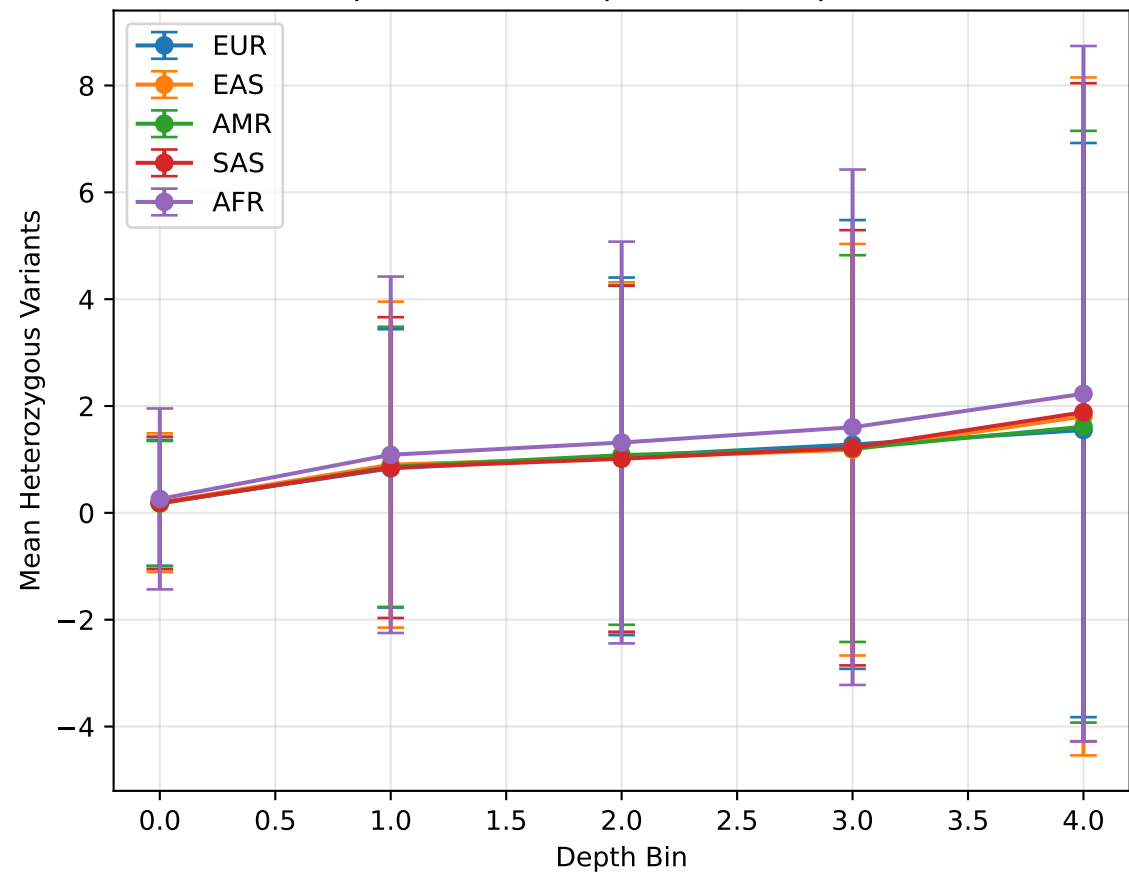
Model 2: All Technical
 $R^2 = 0.0759$



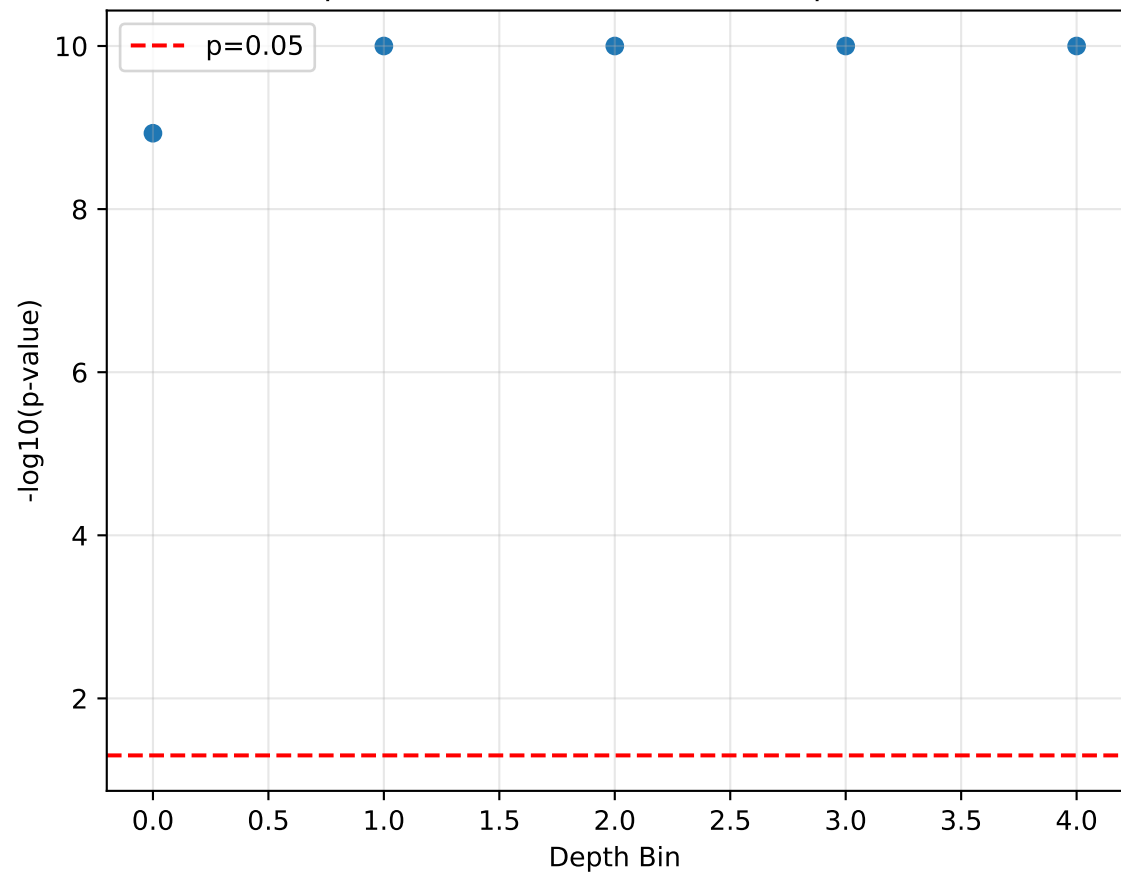
Model 3: Technical + Population
 $R^2 = 0.0772$

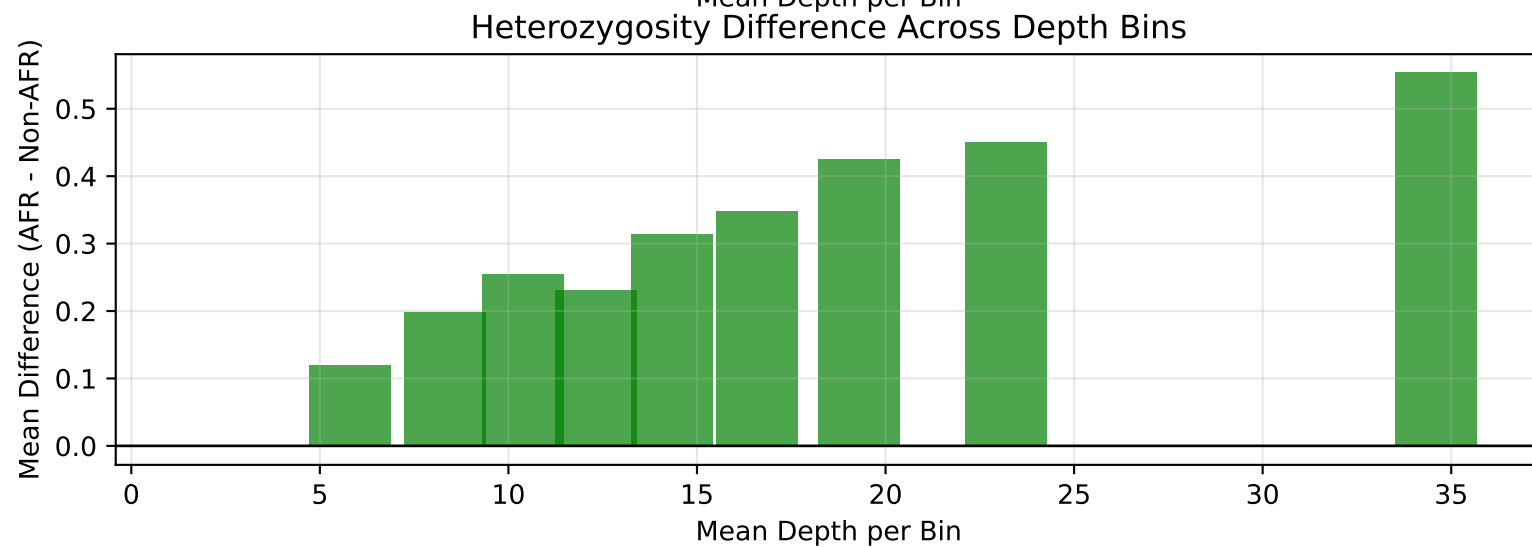
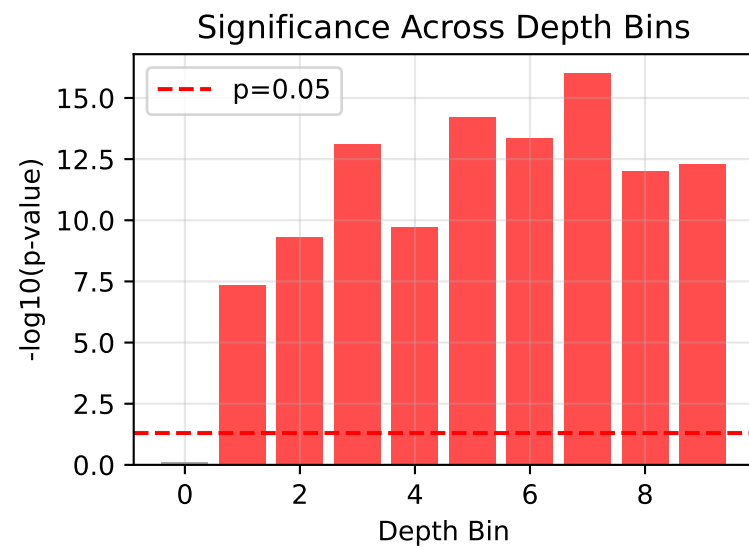
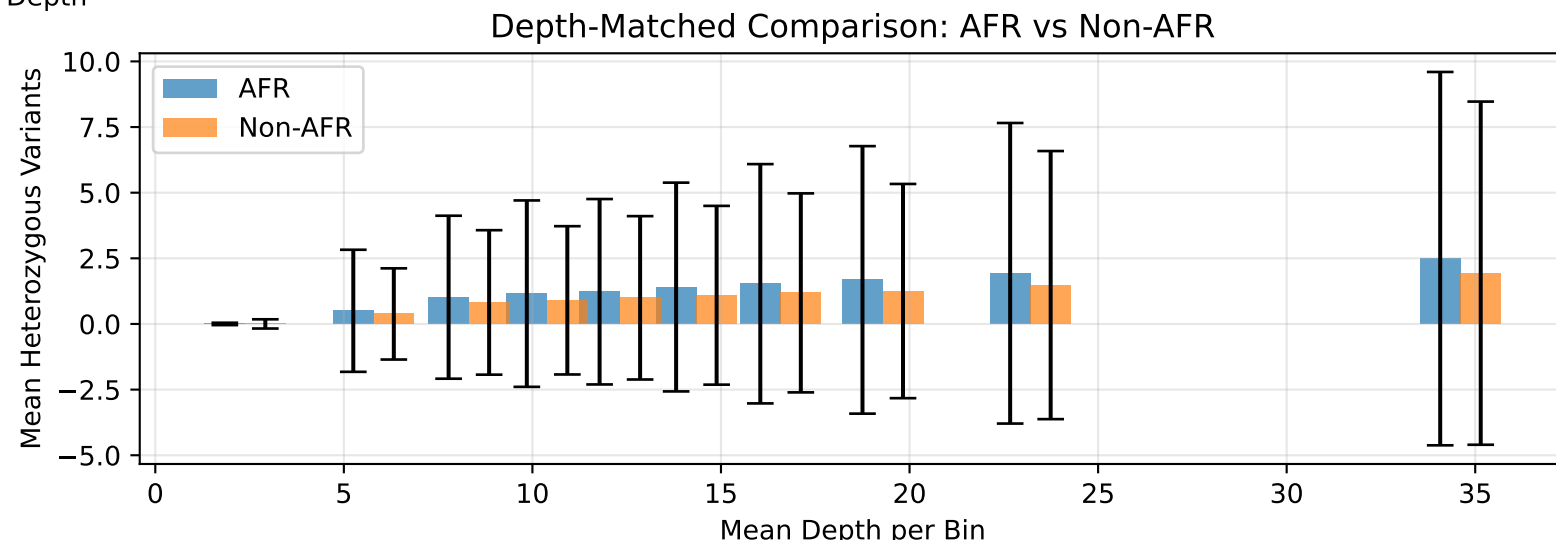
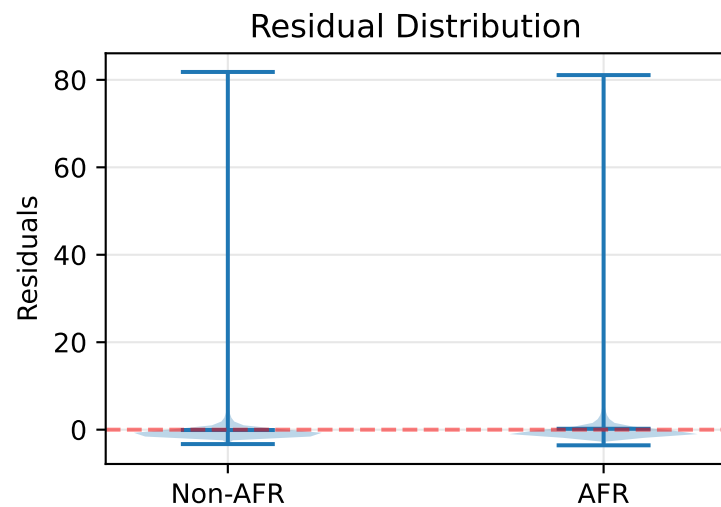
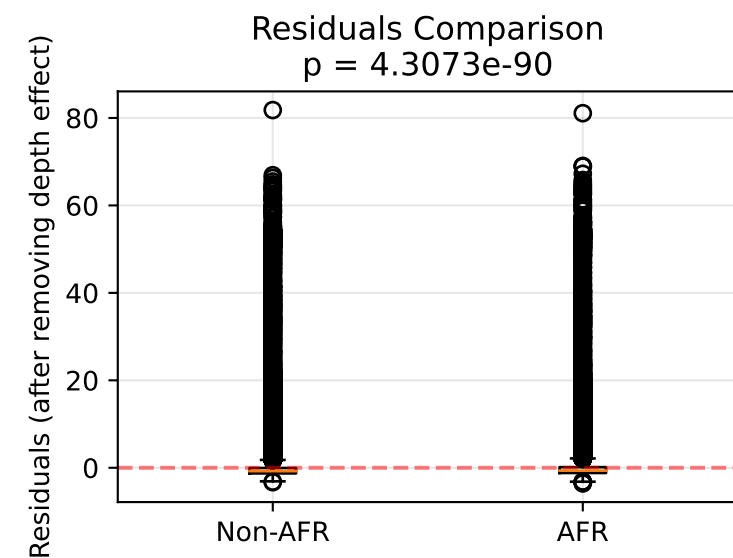
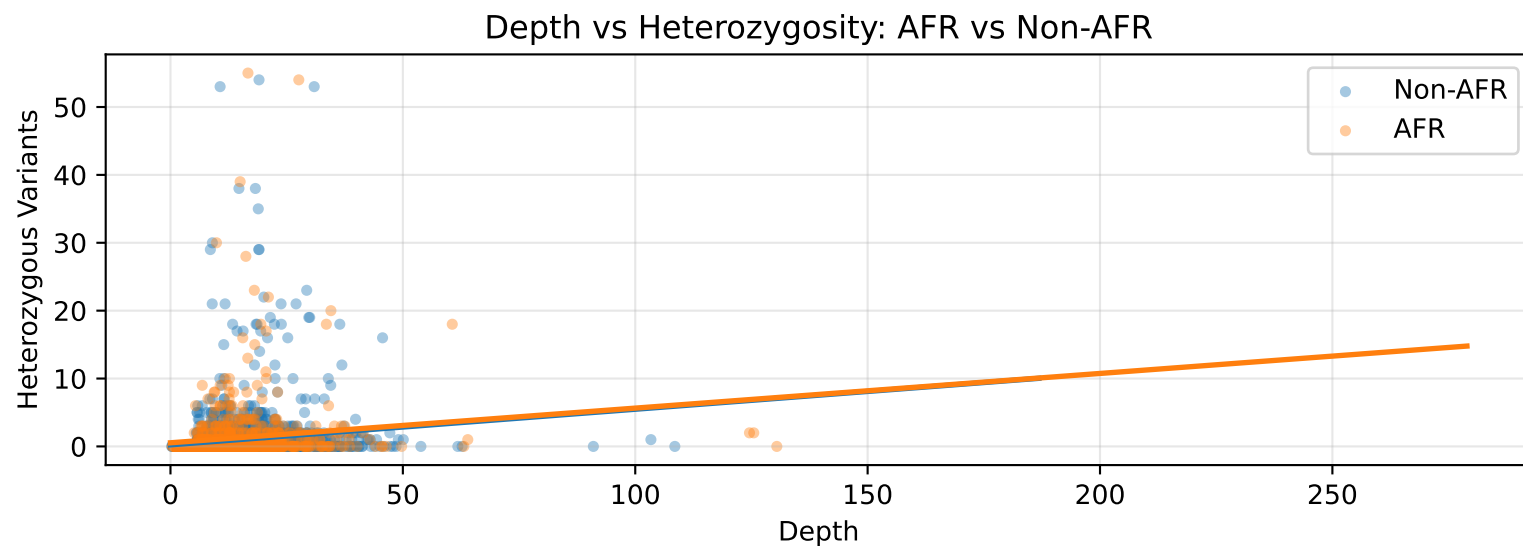


Depth-Matched Population Comparison



Population Differences Within Depth Bins





IG/TCR Heterozygosity Regression Analysis - Summary

DATASET SUMMARY

Total samples: 387,917
Unique genes: 404
Populations: AFR, AMR, EAS, EUR, SAS

Depth Statistics:
Mean: 14.71
Median: 13.36
Range: [0.01, 278.95]

Heterozygous Variant Statistics:
Mean: 1.08
Median: 0.00
Range: [0, 83]

CORRELATION ANALYSIS (Depth vs Heterozygosity)

Pearson correlation: $r = 0.1297$, p-value = $0.0000e+00$
Spearman correlation: $\rho = 0.2432$, p-value = $0.0000e+00$

Interpretation: Weak positive correlation between depth and heterozygosity.

MODEL COMPARISON

Model 1 (Depth only): $R^2 = 0.0141$
Model 2 (All Technical Covariates): $R^2 = 0.0759$
Model 3 (Technical + Population): $R^2 = 0.0772$

Variance explained by technical factors: 7.59%
Additional variance by population: 0.13%

AFR POPULATION ANALYSIS (DEPTH-CONTROLLED)

Question: Does AFR have higher heterozygosity after controlling for depth?

Answer: YES

- Evidence:
- Residual Analysis (depth removed):
 - Difference: 0.2838
 - p-value: $4.3073e-90$
 - Mann-Whitney U Test:
 - p-value: $3.6428e-253$
 - ANCOVA (Depth + AFR indicator):
 - AFR coefficient: 0.2838
 - p-value: $1.1102e-16$
 - Depth-Matched Comparison:
 - Bins where AFR > non-AFR: 9/10
 - Significant bins ($p < 0.05$): 9/10

KEY FINDINGS

- Depth has a significant impact on heterozygous variant detection (Pearson $r=0.130$, $p=0.00e+00$)
- Technical covariates (depth, read length, quality) explain 7.6% of variance in heterozygosity
- Population factors add 0.1% additional explanatory power after controlling for technical factors
- Depth-matching is critical for fair population comparisons
- AFR population DOES show significantly higher heterozygosity after controlling for depth effects

RECOMMENDATIONS

- Use depth-matched comparisons when comparing populations
- Control for technical covariates (depth, read length, quality)
- Consider residual analysis to identify true population effects
- Review depth-matched plots to verify population differences

OUTPUT FILES

All results saved to: regression/

- model_comparison.csv
- population_residuals.csv
- depth_bins_stats.csv
- pop_correlations.csv
- depth_matched_comparison.csv
- depth_distribution_stats.csv
- afr_heterozygosity_summary.csv
- afr_depth_matched_comparison.csv